Part XIII

Principal Component Analysis & Singular Value Decomposition

Lecture Outline

Why Dimension Reduction?

- Data sets often live in *very* high-dimensional spaces.
  - Image data (pixels)
  - Genomics (gene-expression vectors)
  - Text data (token or embedding vectors)

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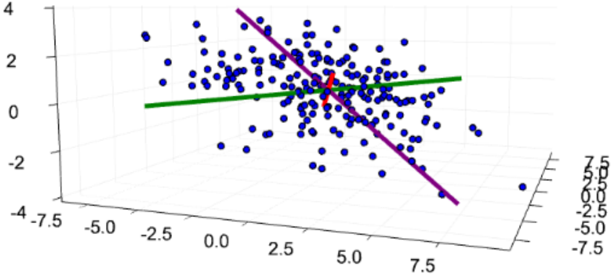
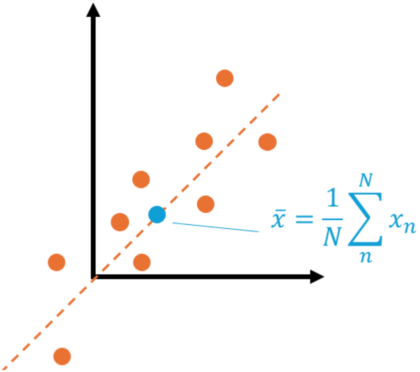
Example

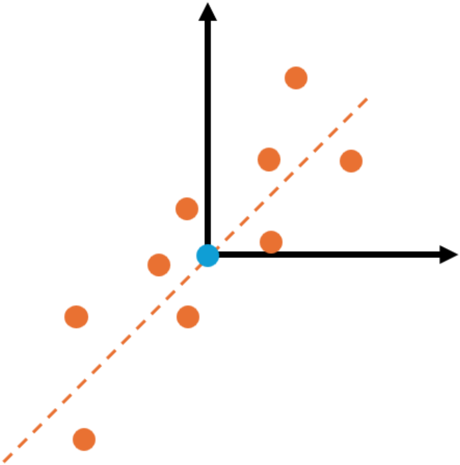
Can you visually see the relevant directions?



- Consider N data points $x_1, \ldots, x_N \in \mathbb{R}^2$.
- Sample mean $\bar{x} = \frac{1}{N}\sum_{n=1}^{N} x_n$.

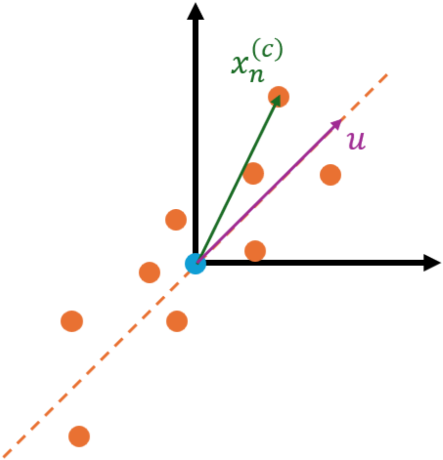


- Centre each sample: $x_n^{(c)} := x_n - \bar{x}, \;\; n \in \{1, \ldots, N\}$.



Projections

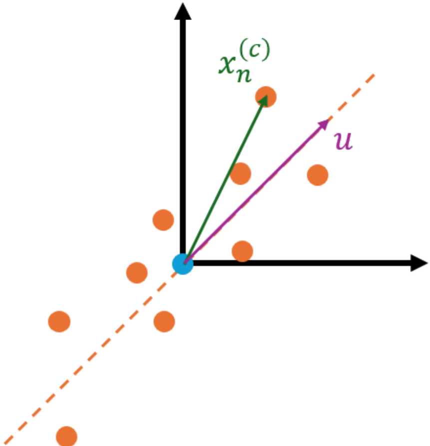
- We now wish to project every $x_n^{(c)}$ onto a unit direction $u \in \mathbb{R}^2$ with $u^\top u = 1$.



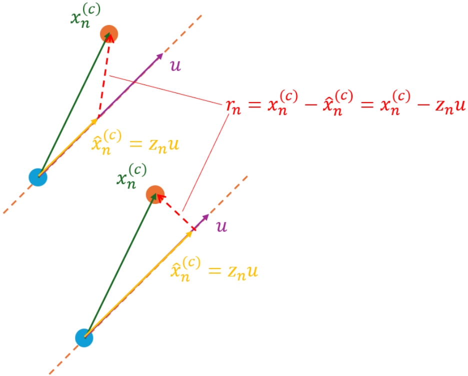
- Decompose

$$x_n^{(c)} = \underbrace{\hat{x}_n^{(c)}}_{\text{parallel to } u} + \underbrace{r_n}_{\text{perpendicular } u},$$

- In other words: $\hat{x}_n^{(c)} = z_n\, u,\ z_n \in \mathbb{R}$ (parallel) and $u^\top r_n = 0$ (perpendicular).



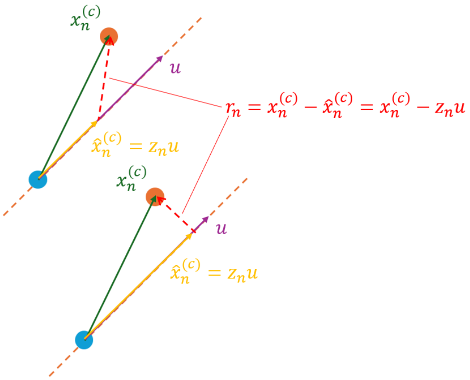
- How to choose z_n?



Projections

- How to choose z_n?
- Minimise the residual's norm:

$$z_n^\star \;=\; \arg\min_{z\in\mathbb{R}} \left\| x_n^{(c)} - z\,u \right\|_2^2 .$$



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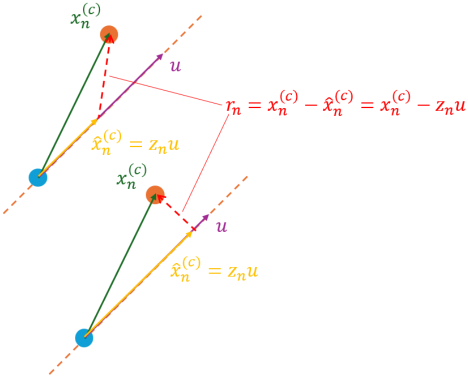
$$z_n^\star \;=\; \arg\min_{z\in\mathbb{R}} \left\| x_n^{(c)} - z\,u \right\|_2^2 .$$

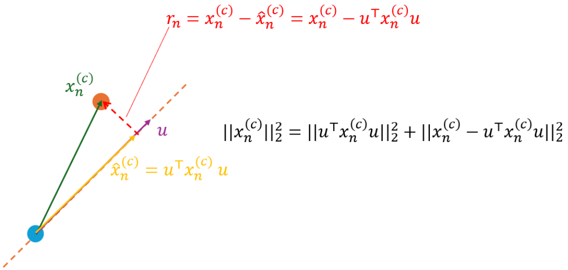
- Least-squares solution:

$$z_n^\star = (u^\top u)^{-1} u^\top x_n^{(c)} = u^\top x_n^{(c)}, \quad \hat{x}_n^{(c)} = (u^\top x_n^{(c)})\,u, \quad r_n = x_n^{(c)} - \hat{x}_n^{(c)}.$$



- Pythagorean identity:

$$\|x_n^{(c)}\|_2^2 = \|\hat{x}_n^{(c)}\|_2^2 + \|r_n\|_2^2 = \|(u^\top x_n^{(c)})\, u\|_2^2 + \|x_n^{(c)} - (u^\top x_n^{(c)})\, u\|_2^2.$$

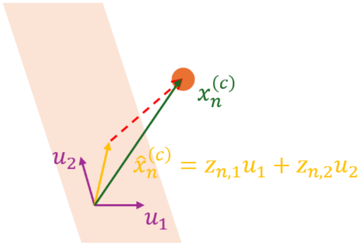


Projections: Higher Dimensions

- Let $x_1, \ldots, x_N \in \mathbb{R}^d$ and pick orthonormal basis vectors $u_1, \ldots, u_K \in \mathbb{R}^d$ obeying

$$u_k^\top u_j = \begin{cases} 1 & k = j, \\ 0 & k \neq j, \end{cases} \qquad k, j \in \{1, \ldots, K\}.$$

Stack them as $U = [u_1 \;\; \ldots \;\; u_K]$ so $U^\top U = I_K$.



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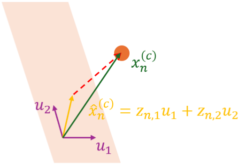
- Decompose each centred sample:

$$x_n^{(c)} = \hat{x}_n^{(c)} + r_n, \qquad \hat{x}_n^{(c)} \in \mathrm{range}(U),\ r_n \perp \mathrm{range}(U).$$

  Because $\hat{x}_n^{(c)} \in \mathrm{range}(U)$, then

$$\hat{x}_n^{(c)} = z_{1,n} u_1 + \dots + z_{K,n} u_K = U z_n,\ z_n = \begin{bmatrix} z_{1,n} \\ \vdots \\ z_{K,n} \end{bmatrix} \in \mathbb{R}^K.$$

  (Compare with $\hat{x}_n^{(c)} = z_n u$ in 1-D.)



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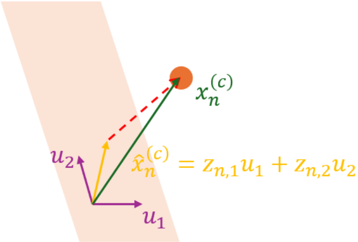
- Least-squares coordinates:

$$z_n^\star = \arg\min_{z \in \mathbb{R}^K} \left\| x_n^{(c)} - Uz \right\|_2^2 \implies z_n^\star = (U^\top U)^{-1} U^\top x_n^{(c)} = U^\top x_n^{(c)}.$$



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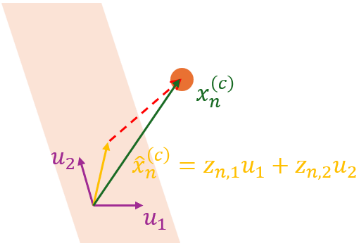
  Stack them as $U = [u_1 \ \ldots \ u_K]$ so $U^\top U = I_K$.
- Hence $\hat{x}_n^{(c)} = Uz_n^\star = UU^\top x_n^{(c)}$, $r_n = (I_d - UU^\top)x_n^{(c)}$. Again

$$\|x_n^{(c)}\|_2^2 = \|z_n^\star\|_2^2 + \|r_n\|_2^2.$$



Projections: Batch View

- Stack all samples:

$$X = [x_1 \ \dots \ x_N] \in \mathbb{R}^{d \times N}, \ X^{(c)} = X - \bar{x}$$

- **Coordinate matrix**

$$Z = U^\top X^{(c)} \in \mathbb{R}^{K \times N},$$

  whose n-th column z_n contains the K coordinates of $x_n^{(c)}$ in the basis $\{u_1, \dots, u_K\}$.

- **Reconstruction in subspace**

$$\hat{X} = \hat{X}^{(c)} + \bar{x} = UZ + \bar{x} = UU^\top X^{(c)} + \bar{x}.$$

- **Residuals**

$$X - \hat{X} = X^{(c)} - \hat{X}^{(c)} = (I_d - UU^\top)X^{(c)}.$$

In machine learning, Z is called the *representation* or *code* or *scores* of X while $\hat{X}$ is called the *reconstruction* of X. The individual components (elements) of each vector $u_1, \dots, u_K$ are called *feature loadings*.

From Reconstruction Error to Variance

- Goal: find a unit direction u that minimizes the average **residual** (average reconstruction error)

$$E(u) \;=\; \frac{1}{N}\sum_{n=1}^{N} \|x_n^{(c)} - (u^\top x_n^{(c)})\, u\|_2^2$$

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Therefore

$$E(u) \;=\; \frac{1}{N}\sum_{n=1}^{N} \|x_n^{(c)}\|_2^2 - \underbrace{\frac{1}{N}\sum_{n=1}^{N} (u^\top x_n^{(c)})^2}_{\text{projected variance } (\mathrm{Var}(z_n))} .$$

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  Therefore

- **Minimising** $E(u) \iff$ **maximising the captured variance** $\frac{1}{N}\sum_{n=1}^{N}(u^\top x_n^{(c)})^2.$

Sample Covariance Matrix

- Note that

$$\frac{1}{N}\sum_{n=1}^{N}(u^\top x_n^{(c)})^2 = \frac{1}{N}\sum_{n=1}^{N}(u^\top x_n^{(c)})((x_n^{(c)})^\top u)$$

$$= \frac{1}{N}u^\top(\sum_{n=1}^{N}x_n^{(c)}(x_n^{(c)})^\top)u = u^\top\left(\frac{1}{N}X^{(c)}\left(X^{(c)}\right)^\top\right)u.$$

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So we want to find the unit vector u such that we maximize $u^\top Cu$.

Properties of the Covariance Matrix C

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- **Symmetric**: $C = C^\top$.
- **Positive semi-definite** (PSD): for every v, $v^\top C v = \|(X^{(c)})^\top v\|_2^2 / N \geq 0$.
- Consequences:
  - Real, non-negative eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$.
  - Eigenvectors for distinct λ_i are orthogonal ($v_i^\top v_j = 0, \; i \neq j$).

Deriving PCA with optimization: first direction u_1

- We set up the constrained problem:

$$\max_u \; u^\top C \, u \quad \text{s.t.} \;\; u^\top u = 1.$$

- Lagrangian $\mathcal{L}(u, \lambda) = u^\top Cu - \lambda(u^\top u - 1)$.
- Stationary point: $Cu = \lambda u$

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- This means that u is an eigenvector of C with eigenvalue λ. Note also that $u^\top Cu = \lambda$.
- We want to maximize $u^\top Cu$, so we choose the largest eigenvalue of C: $\lambda = \lambda_1$.

$$\boxed{u_1 = \text{eigenvector of } C \text{ for } \lambda_1}$$

Deriving PCA with optimization: Second Direction u_2

- For the second direction, we add an orthogonality constraint:

$$\max_{u_2} \; u_2^\top C u_2 \;\; \text{s.t. } u_2^\top u_2 = 1, \; u_2^\top u_1 = 0.$$

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- Optimum obeys $Cu_2 = \lambda u_2 \Rightarrow u_2$ is another eigenvector.
- Pick the next-largest eigenvalue λ_2.

Deriving PCA with optimization: general k-th direction u_k

- For the k-th direction, we have the constraints: $\|u_k\| = 1$ and $u_k^\top u_j = 0$ for $j < k$.

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Deriving PCA with optimization: general k-th direction u_k

- For the k-th direction, we have the constraints: $\|u_k\| = 1$ and $u_k^\top u_j = 0$ for $j < k$.
- Same Lagrange-multiplier trick $\Rightarrow Cu_k = \lambda_k u_k$.
- Choose eigenvector associated with the k-th largest eigenvalue λ_k.

Deriving PCA with optimization: Summary

- We choose the directions $u_1, \ldots, u_K$ as the eigenvectors of C corresponding to largest K eigenvalues $\lambda_1, \ldots, \lambda_K$ of C.
- These directions are the Principal Components (PCs).

Eigenvalue Decomposition & PCA Coordinate Transform

Diagonalizable matrix. Let $A \in \mathbb{R}^{d\times d}$ (square matrix). We call A *diagonalizable* if there exists an invertable $Q \in \mathbb{R}^{d\times d}$ and a diagonal matrix $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$ such that

$$A = Q\,\Lambda\,Q^{-1}.$$

Here $\lambda_1, \dots, \lambda_d$ are the eigenvalues of A, and the columns $q_1, \dots, q_d$ of Q are the corresponding eigenvectors.

Eigenvalue Decomposition & PCA Coordinate Transform

Symmetric case. If $A = A^\top$, then one can choose an orthonormal eigenbasis:

$$A = Q\,\Lambda\,Q^\top, \quad Q^\top Q = I_d \quad (Q^{-1} = Q^\top), \quad \Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d),\ \lambda_1, \dots, \lambda_d \in \mathbb{R}.$$

In particular, for the covariance C, all $\lambda_1, \dots, \lambda_d \geq 0$ since C is PSD. Usually, the order of them is chosen such that $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$.

Eigenvalue Decomposition & PCA Coordinate Transform

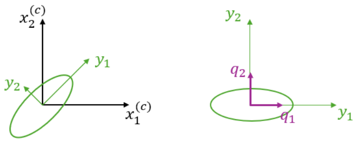
Since $Q^\top Q = I$, the matrix $Q^\top$ is an orthonormal transformation (a rotation). For the centred data matrix $X^{(c)}$, define

$$Y = Q^\top X^{(c)}.$$

Then the covariance of Y is

$$\mathrm{Cov}[Y] = Q^\top C \, Q = \Lambda,$$

which is diagonal. Thus, the rotation by $Q^\top$ aligns the coordinate axes with the principal directions, making the new coordinates uncorrelated.



Eigenvalue Decomposition & PCA Coordinate Transform

Let $K \leq d$. Take the top-K eigenpairs $(\lambda_i,\, q_i)$ for $i = 1, \ldots, K$ (largest $\lambda_1, \ldots, \lambda_K$). Stack

$$U = [\, q_1 \;\; \ldots \;\; q_K \,] \in \mathbb{R}^{d \times K}, \quad U^\top U = I_K.$$

Eigenvalue Decomposition & PCA Coordinate Transform

For the centred data matrix $X^{(c)}$:

$$Z = U^\top X^{(c)} \in \mathbb{R}^K, \quad \hat{X}^{(c)} = U\, Z,$$

In the $\{q_i\}$-basis, $\mathrm{Cov}[Z] = \mathrm{diag}(\lambda_1, \ldots, \lambda_K)$, *so each coordinate variance equals its eigenvalue*. These coordinates are *uncorrelated*.

Explained Variance

- How should we choose K?
- If we use K principal components, then to measure how much they capture the variance, we use the *Individual Explained Variance* for PC_i (u_i):

$$\frac{\lambda_i}{\sum_{i=1}^{d} \lambda_i},$$

or the *Cumulative Explained Variance*:

$$\frac{\sum_{i=1}^{K} \lambda_i}{\sum_{i=1}^{d} \lambda_i}.$$

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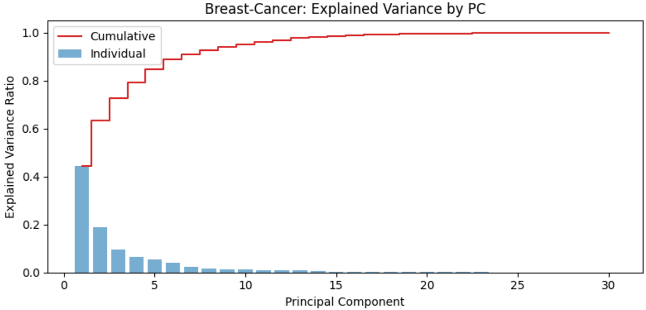

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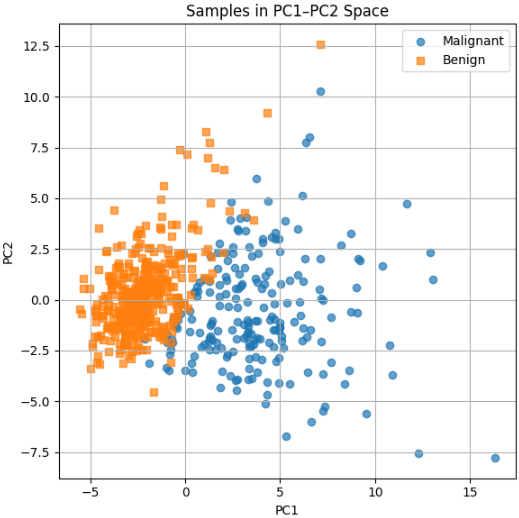

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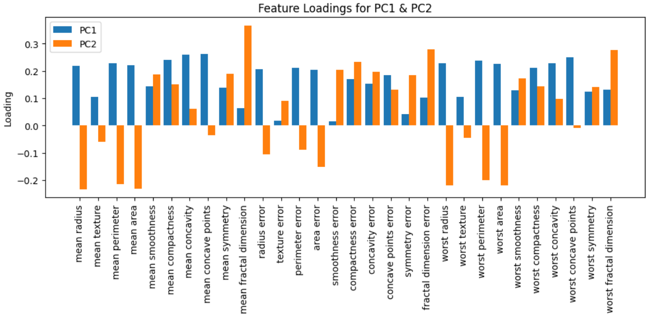

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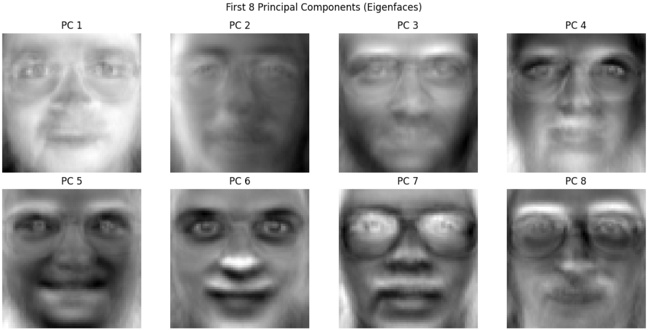

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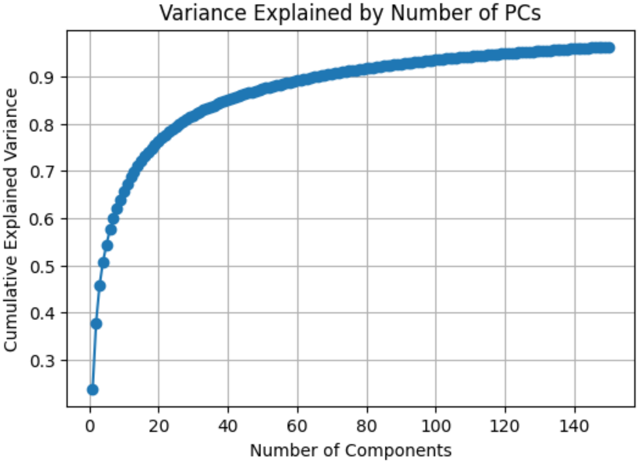

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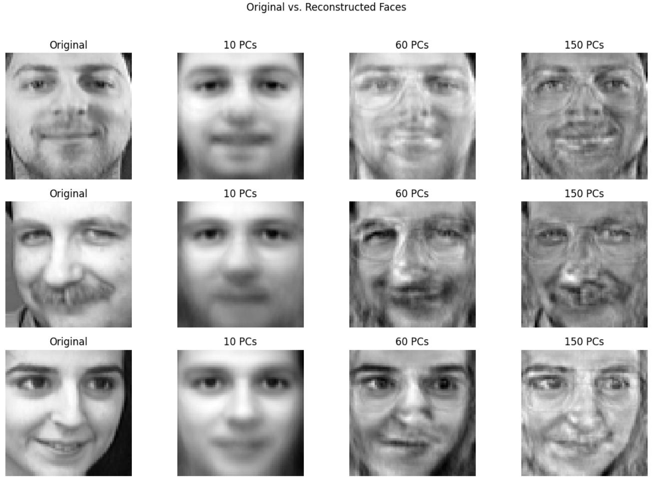

Numerical Challenges with EVD

- **Problem 1**: Explicitly computing $C = \frac{1}{N} X^{(c)} (X^{(c)})^\top$ and storing it is *computationally expensive* for high-dimensional data ($d \gg 1$).
- **Problem 2**: EVD on C can be susbtible to numerical instabilities.
- **Solution**: Use **Singular Value Decomposition (SVD)** directly on $X^{(c)}$. Avoids forming C and is more numerically stable.

Singular Value Decomposition: Definition

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with $U^\top U = I_d$, $V^\top V = I_N$, and a diagonal matrix

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & 0 \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_r & \\ 0 & & & & 0 \end{bmatrix} \in \mathbb{R}^{d \times N}, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0,$$

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- Equivalently, in sum-of-rank-one form:

$$X = \sum_{i=1}^{r} \sigma_i \, u_i \, v_i^\top,$$

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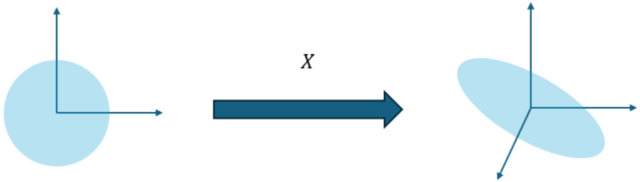
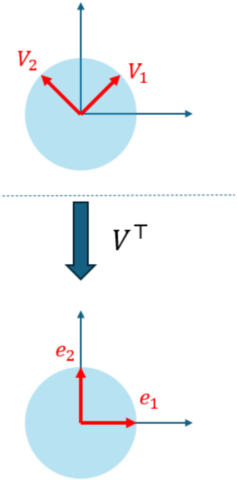
where $r = \mathrm{rank}(X)$ and each term $\sigma_i \, u_i \, v_i^\top$ is a rank-one matrix.
- The non-negative scalars $\{\sigma_i\}$ are the *singular values* of X.

- The image of the unit sphere under any $d \times N$ matrix X is a hypersphere



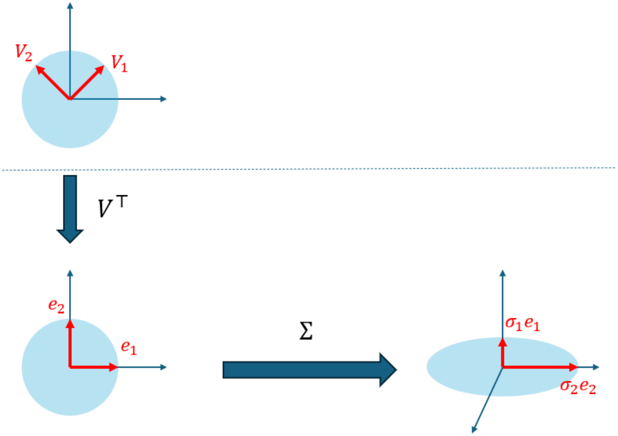
Geometric Interpretation

- We first rotate the unit sphere by $V^\top$

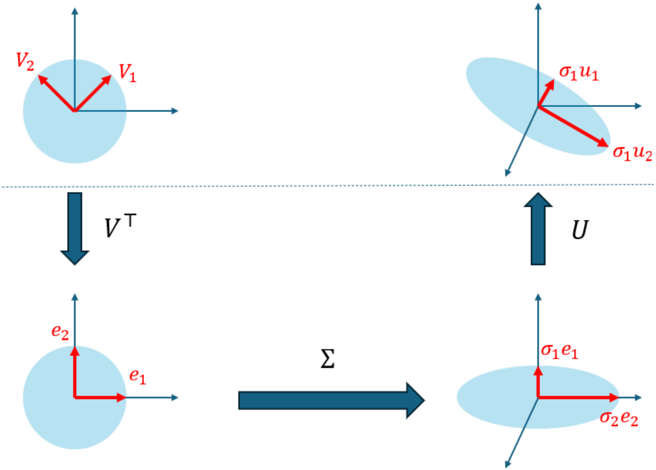


Geometric Interpretation

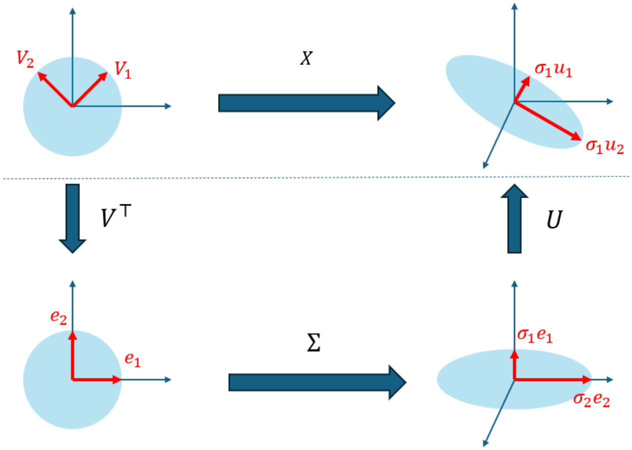
- Then we scale by Σ



- Then we rotate again by U



- Summary.


V_2
V_1
X
$\sigma_1 u_1$
$\sigma_1 u_2$
$V^\top$
U
e_2
e_1
Σ
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Rank, Null Space, and Range via SVD

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- The conditioning number of a matrix is the ratio of the largest to the smallest singular value.
- SVD can be extended to complex matrices if $\top$ is understood as the transpose of the conjugate matrix.

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omitting all zero-singular-value directions.
- Storage and computation comparison:

$$\widetilde{U}: \; d \times r, \quad \widetilde{\Sigma}: \; r \times r, \quad \widetilde{V}: \; N \times r \quad \text{vs.} \quad U: \; d \times d, \; \Sigma: \; d \times N, \; V: \; N \times N.$$

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Remember the EVD $C = Q \Lambda Q^\top$.

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- Hence the eigenvalues of C are $\lambda_i = \sigma_i^2 / N$, with eigenvectors u_i. Equivalently, $C\, u_i = (\sigma_i^2 / N)\, u_i$.

Using SVD to Compute PCA

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  so row i of Z has variance σ_i^2 / N.
- Truncating to the top K singular values gives exactly the best rank-K PCA approximation.

Truncated SVD for Dimensionality Reduction

- If we find U, Σ, and V in $X^{(c)}$, then we can obtain an approximate

$$\hat{X}^{(c)} = U_K \Sigma_K V_k,$$

where $U_K \in \mathbb{R}^{d \times K}$, $\Sigma_K \in \mathbb{R}^{K \times K}$, $V_K \in \mathbb{R}^{N \times K}$ are chosen based on the K top singular values.

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- $\hat{X}^{(c)} = U_K \Sigma_K V_K$ is exactly the same projection onto the first K components $\hat{X}^{(c)} = U_K U_K^\top X^{(c)}$
- To see that, note:

$$X^{(c)} = U \Sigma V^\top \Longrightarrow U_K^\top X^{(c)} = U_K^\top (U \Sigma V^\top) = \underbrace{(U_K^\top U)}_{[I_K \;\; 0]} \Sigma V^\top = \Sigma_K V_K^\top.$$

Hence, $\hat{X}^{(c)} = U_K U_K^\top X^{(c)} = U_K \Sigma_K V_K$

Advantages of SVD for PCA

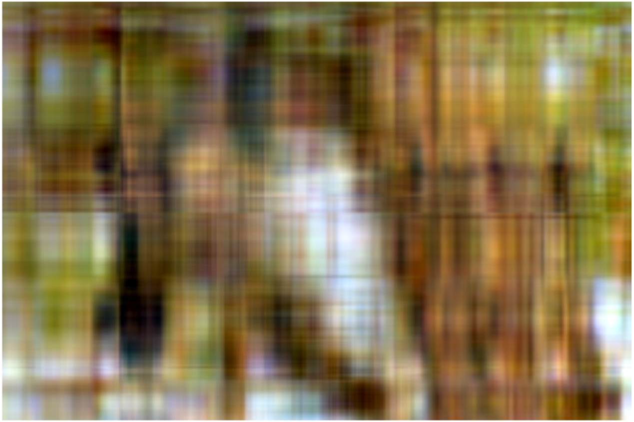
- Instead of computing and storing C, we can obtain the principle components directly from $X^{(c)}$.
- More numerically stable than the EVD with C (handles rank deficiency problems better).

Example: Image Compression

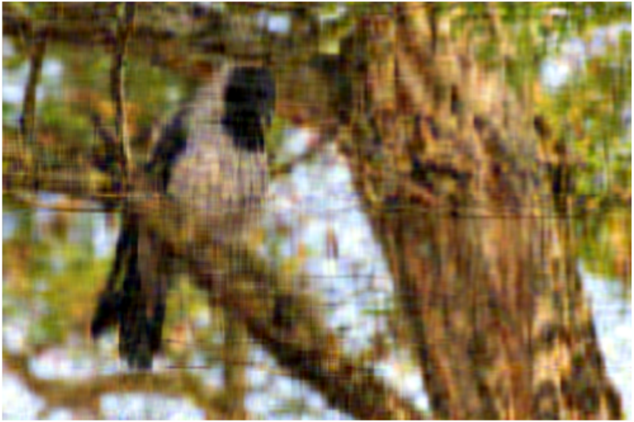
Original Image

Example: Image Compression

Rank-5 Approximation



Rank-20 Approximation



Example: Image Compression

Rank-100 Approximation

Summary: PCA & SVD

- **PCA Goal:** Find orthonormal directions $u_1, \ldots, u_K$ that maximize projected variance (equivalently minimize reconstruction error).
  - Compute sample covariance $C = \frac{1}{N} X^{(c)} (X^{(c)})^\top$.
  - Solve $C\, u_i = \lambda_i u_i$; take top K eigenvectors.
  - Coordinates $Z = U^\top X^{(c)}$, with $\mathrm{Cov}[Z] = \mathrm{diag}(\lambda_1, \ldots, \lambda_K)$.
- **SVD Connection:**

$$X^{(c)} = U \Sigma V^\top \quad \Longrightarrow \quad C = \frac{1}{N}\, U \Sigma^2 U^\top,$$

  so u_i are the left-singular vectors, and $\lambda_i = \sigma_i^2 / N$.
- **Truncated SVD** gives best rank-K approximation:

$$\hat{X}^{(c)} = U_K\, \Sigma_K\, V_K^\top \;=\; U_K U_K^\top X^{(c)}.$$

- **Benefits of SVD for PCA:**
  - Avoid forming C explicitly (storage & speed).
  - More numerically stable (handles high-dim & rank-deficiency).